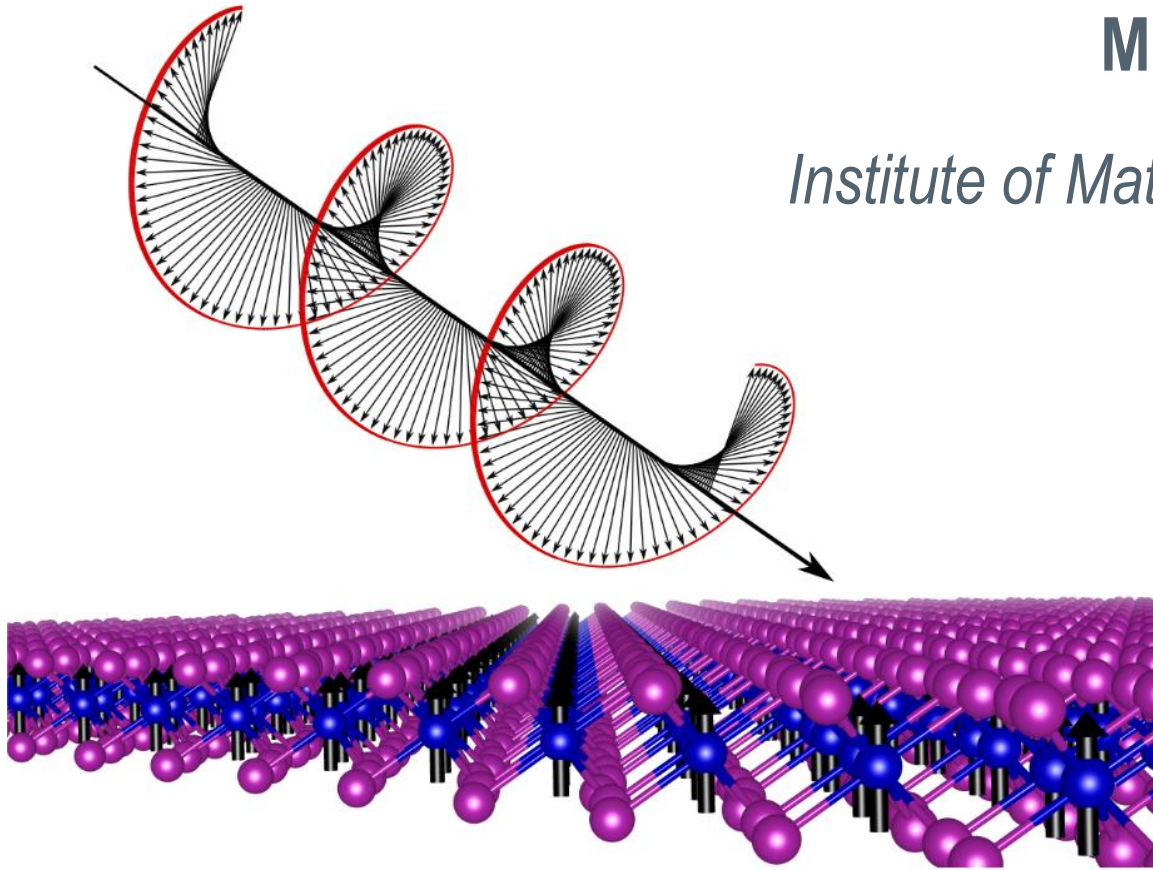


Linear Response of Magnets from Constrained DFPT

Miquel Royo and Massimiliano Stengel

Institute of Materials Science of Barcelona (ICMAB-CSIC)



ANYS

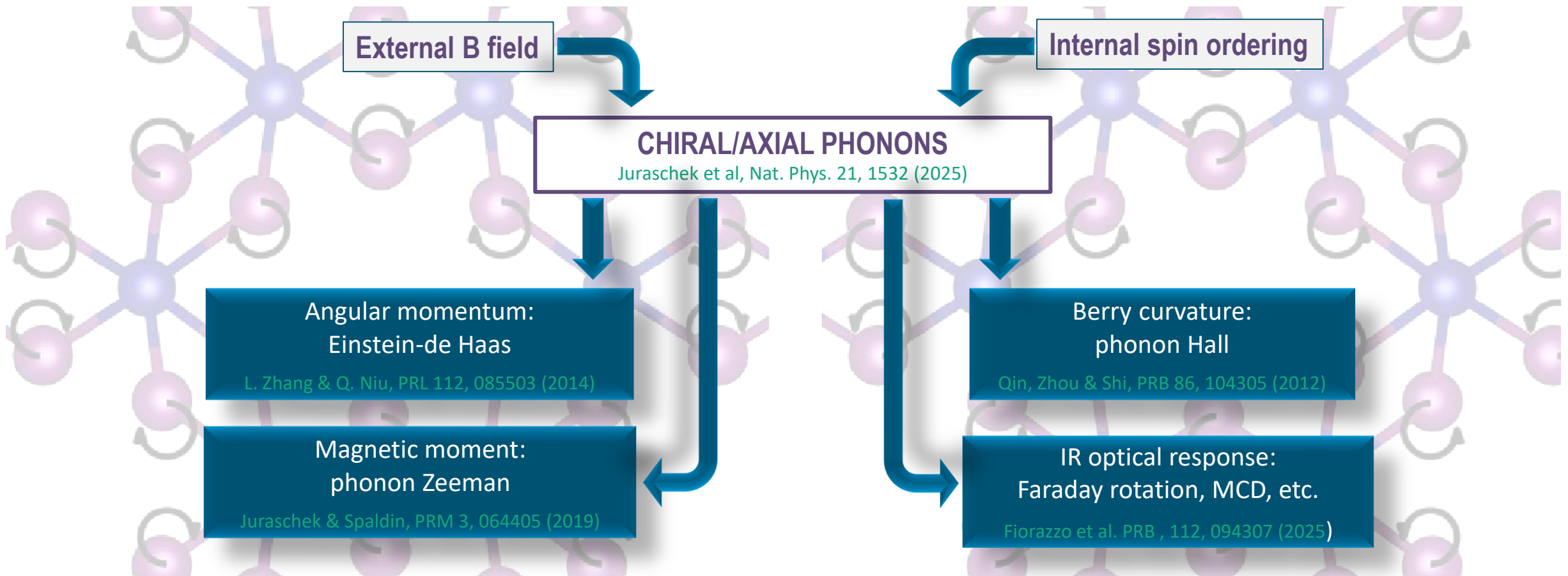
ICMAB

INSTITUT DE CIÈNCIA DE MATERIALS DE BARCELONA



EXCELENCIA
SEVERO
OCHOA

Chiral Phonons via Magnetism



QUESTION: Are standard first-principles phonon calculations capable of capturing these effects?

Born-Oppenheimer Dynamics

In a *strict* adiabatic (Born-Oppenheimer) approximation, electrons adapt instantaneously to any change due to a perturbation (atomic displacements).

Force-constants matrix

$$C_{\kappa\alpha,\kappa'\beta}^{\mathbf{q}} = \frac{\partial^2 E}{\partial \tau_{\kappa\alpha}^{-\mathbf{q}} \partial \tau_{\kappa'\beta}^{\mathbf{q}}}$$

κ, κ' : sublattice indices α, β : Cartesian directions



Equation of motion

$$\omega^2 \mathbf{M}|\eta\rangle = \mathbf{C}|\eta\rangle$$

✓ Accurate $C_{\kappa\alpha,\kappa'\beta}^{\mathbf{q}}$ at any $\mathbf{q}$ point from first principles via DFPT

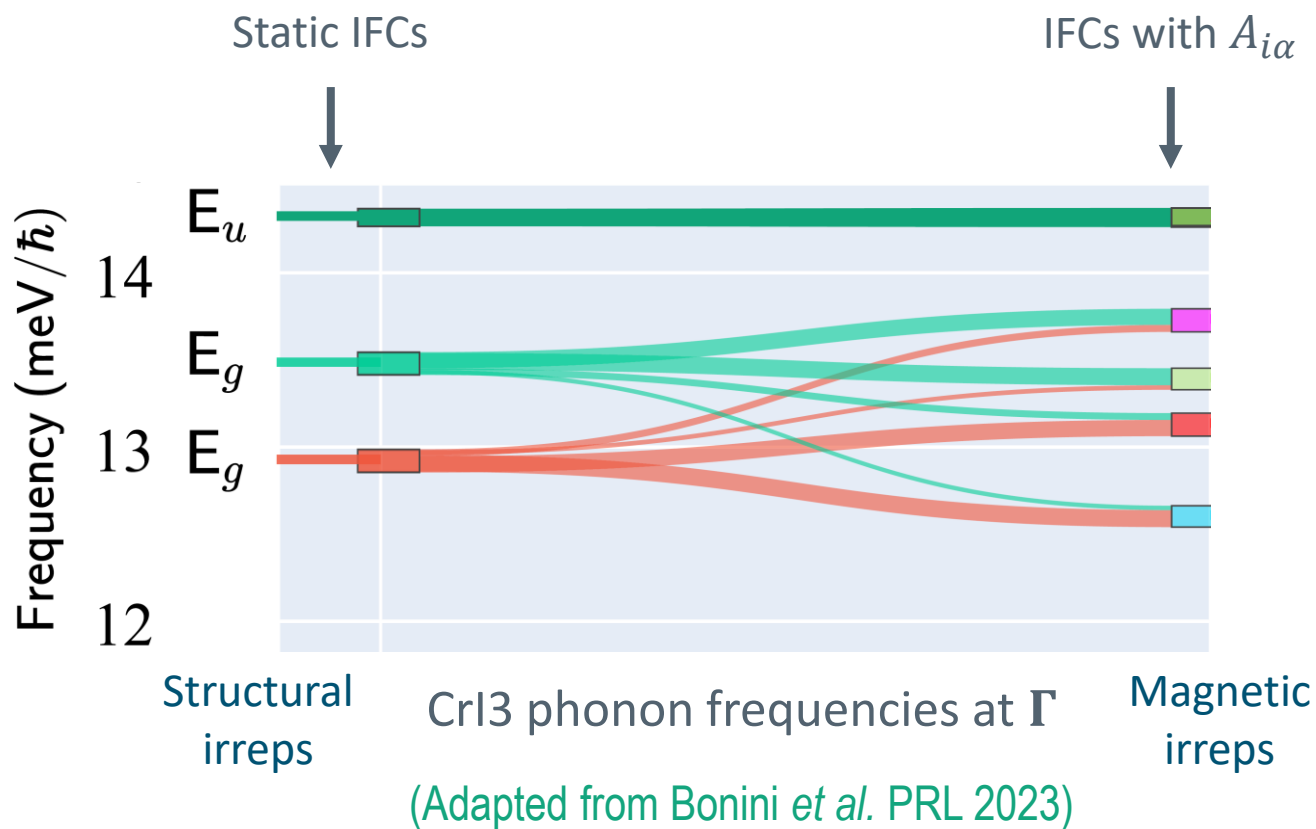
X. Gonze PRB **55**, 10337 (1997), X. Gonze and C. Lee PRB **55**, 10355 (1997), S. Baroni *et al.*, RMP **73**, 515 (2001)

Calculated force constants are static: phonons do not break TR symmetry

$$C_{\kappa\alpha,\kappa'\beta}^{-\mathbf{q}} = \left(C_{\kappa\alpha,\kappa'\beta}^{\mathbf{q}} \right)^*$$

Adiabatic Dynamics in Magnetic Insulators

Standard treatment: broken TRS is not transferred to the response



Adiabatic Lattice Dynamics

Mead & Truhlar JCP(1979)

$$H_{\text{eff}} = \sum_{i\alpha} \frac{[p_{i\alpha} - \hbar A_{i\alpha}(\mathbf{R})]^2}{2m_i} + V_{\text{eff}}(\mathbf{R})$$

$$\omega_n^2 \mathbf{M} \eta_n = (\mathbf{C} + i\omega_n \mathbf{G}) \eta_n$$

$A_{i\alpha}$: Nuclear Berry potential

$\mathbf{G}$: Nuclear Berry curvature

Vanishing effect in $\mathcal{T}$ -symmetric systems

Adiabatic Dynamics in Magnetic Insulators

In a *strict* adiabatic (Born-Oppenheimer) approximation, electrons adapt instantaneously to any change due to a perturbation (atomic displacements).

Valid in magnetic insulators?

0.1 to few eV

Electronic excitations
(fast variables)

few to tens THz

Phonon and magnon
(slow variables)

Coupled phonon-magnon **adiabatic** dynamics

Ren, Bonini, Dreyer, Stengel, Vanderbilt, PRX 2024

$$-\omega_n^2 M |q_n\rangle = -K |q_n\rangle - i\omega_n G |q_n\rangle$$

$$M = \begin{pmatrix} M^{(pp)} & 0 \\ 0 & 0 \end{pmatrix} \quad K = \begin{pmatrix} K^{(pp)} & K^{(ps)} \\ K^{(sp)} & K^{(ss)} \end{pmatrix} \quad G = \begin{pmatrix} G^{(pp)} & G^{(ps)} \\ G^{(sp)} & G^{(ss)} \end{pmatrix}$$

Strong renormalization of zone-center axial-phonon splitting
due to magnon coupling.

Nonadiabatic Dynamics: Frequency-Dependent DFPT

Perturbation at finite frequency

$$\Delta V(\mathbf{r}, t) = \lambda_i^{\mathbf{q}, \omega} \varphi_i^q(\mathbf{r}) e^{-i(\mathbf{q} \cdot \mathbf{r} + \omega t) + \eta t}$$

Second derivatives in perturbations

$$U_{ab}(\omega, \mathbf{q}) = \frac{\partial^2 U}{\partial \lambda_a^{-\mathbf{q}, -\omega} \partial \lambda_b^{\mathbf{q}, \omega}}$$

Kohn-Sham “internal action” functional

Dal Corso et al. PRB 1996

$$U_{\text{KS}}(\lambda, \psi) = \frac{1}{T} \int_0^T dt \left[-i \int [d^3 k] \sum_m \langle \psi_{n\mathbf{k}}(t) | \dot{\psi}_{n\mathbf{k}}(t) \rangle \right. \\ \left. + E_{\text{KS}}(t) - \sum_{\kappa\alpha} \frac{M_\kappa}{2} \dot{\tau}_{\kappa\alpha}^2 \right],$$

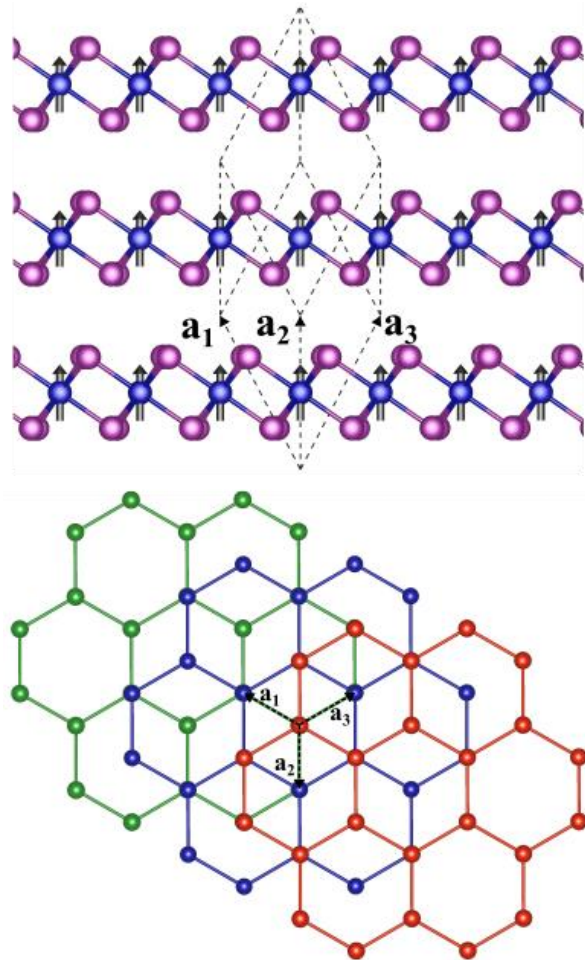
	τ	$\mathcal{E}$	$\mathbf{H}$
Atomic displacement $\rightarrow \tau$	Φ		
Electric field $\rightarrow \mathcal{E}$	$\mathbf{Z}^*$	ϵ	
Zeeman field $\rightarrow \mathbf{H}$ (Macroscopic and local)	$\mathbf{M}^*$	α	χ

Starting point: existing implementation of the finite- $\mathbf{q}$ noncollinear static response in ABINIT

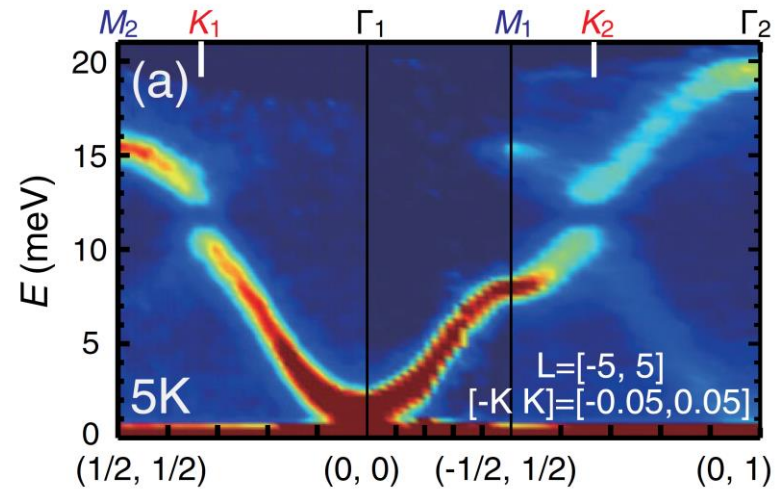
F. Ricci et al., Phys. Rev. B 99, 184404 (2019)

Testcase: Bulk CrI_3

Hexagonal vdW insulating ferromagnet



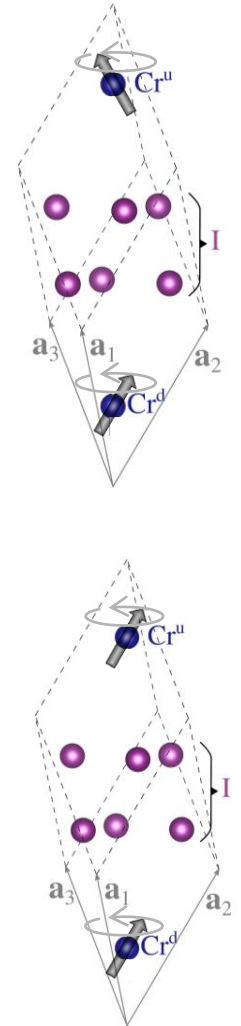
Zone-center magnons



Chen et al. PRX 2021

Optical Magnon (E_u)
 $E_{exp} \approx 19 \text{ meV}$

Acoustic Magnon (E_g)
 $E_{exp} \approx 0.3 \text{ meV}$

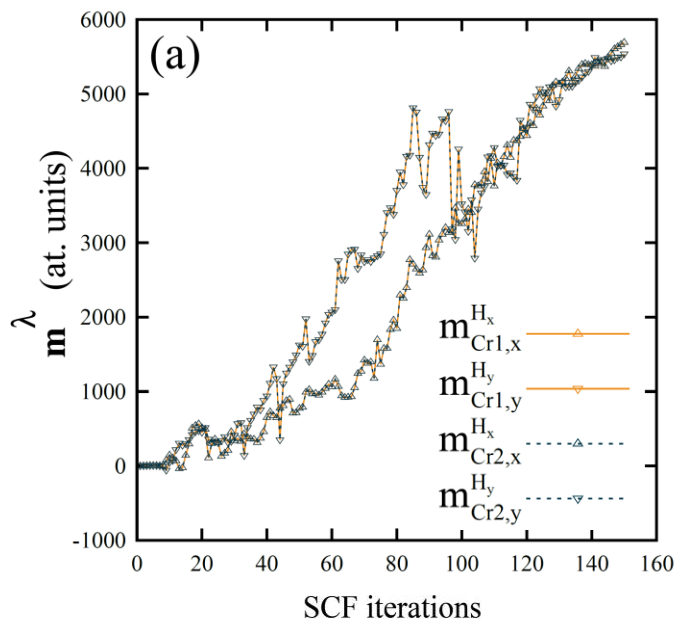


Dynamic DFPT in practice

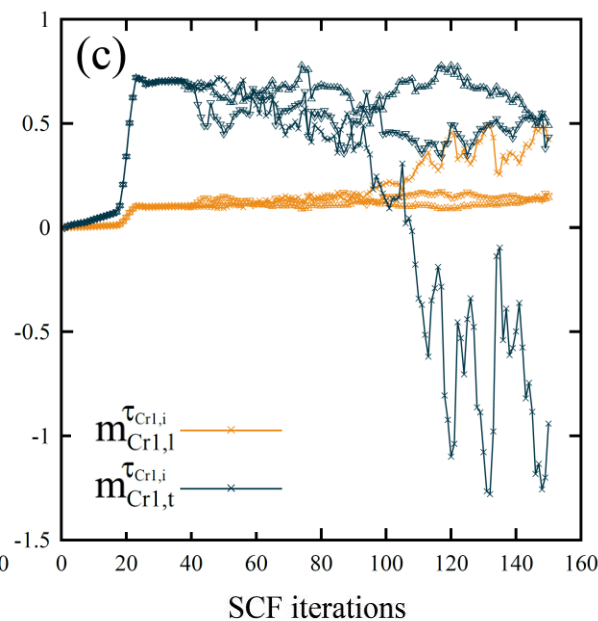
Convergence rate of induced local magnetic moments
(at zone center, $q=0$)

$$\mathbf{m}_{\kappa}^{\lambda_i} = \int d^3r \mathbf{m}^{\lambda_i}(\mathbf{r}) f_{\kappa}(\mathbf{r} - \boldsymbol{\tau}_{\kappa})$$

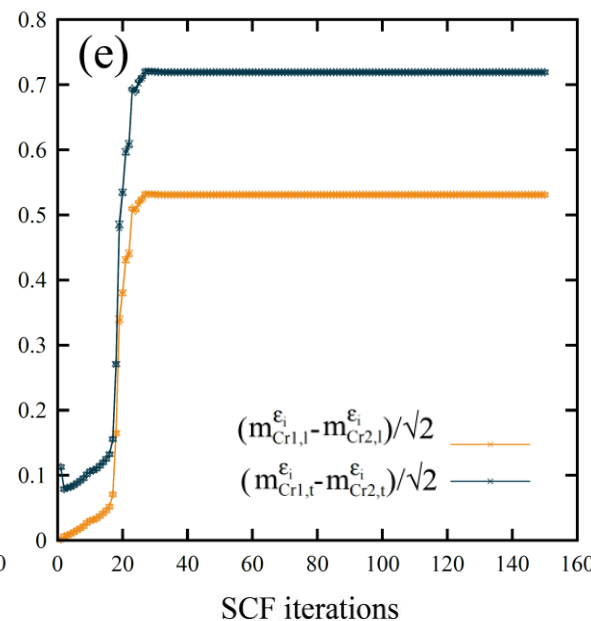
Zeeman ($\hbar\omega = 0$)



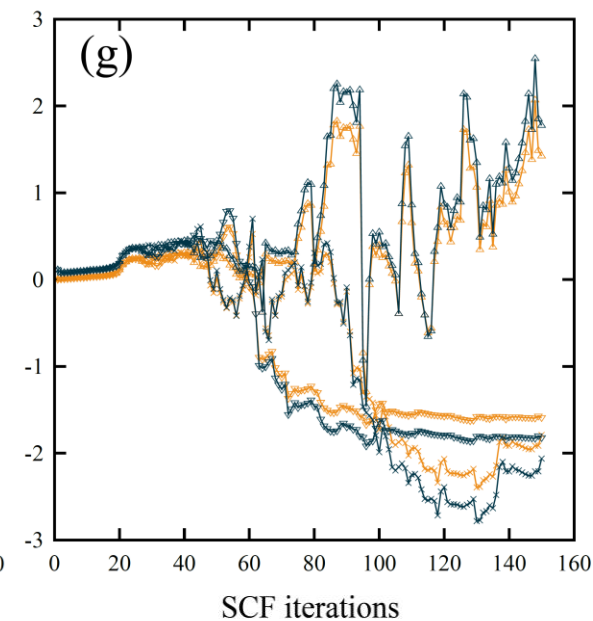
Atomic disp. ($\hbar\omega = 0$)



E field ($\hbar\omega = 0$)



E field ($\hbar\omega \approx 33 \text{ meV}$)



Low-lying magnon modes make the condition number to diverge $\sim 1/\omega_m$

DFPT at Constrained Magnetic Moments

- Ground-state formulation by P.W. Ma and S.L. Dudarev (Phys. Rev. B 91, 054420 (2015))

Penalty amplitude

$$\tilde{U}(\psi, \lambda, B_l) = E_{\text{KS}}(\psi, \lambda) + \frac{\alpha}{2} \sum_l [B_l - m_l(\psi, \lambda)]^2$$

ψ : spinor wave functions

B_l : target local moments

λ : perturbation parameters

l : magnetic degrees of freedom

$$\mathbf{m}_\kappa = \int_{\Omega(\kappa)} d^3r \mathbf{m}(\mathbf{r})$$

atomic spheres spin density

- Here, we extend its application to the linear-response regime

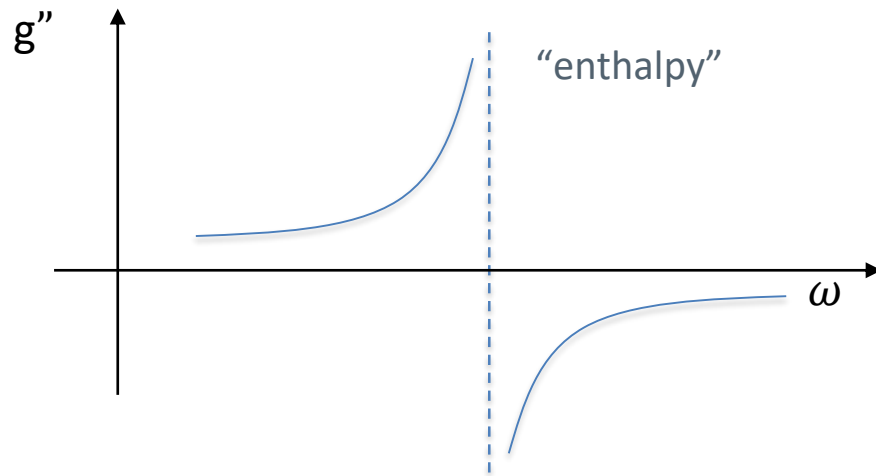
$$\tilde{U}^{\lambda_1 \lambda_2}(\mathbf{q}, \omega) = E^{\lambda_1 \lambda_2}(\mathbf{q}, \omega) + \frac{\alpha}{2} \sum_l [B_l^{\lambda_1} - m_l^{\lambda_1}(\mathbf{q}, \omega)]^* [B_l^{\lambda_2} - m_l^{\lambda_2}(\mathbf{q}, \omega)]$$

- Penalty stiffens the spin fluctuations, removing the problematic low-lying modes
- Second derivatives of the constrained functional are unphysical: what to do?

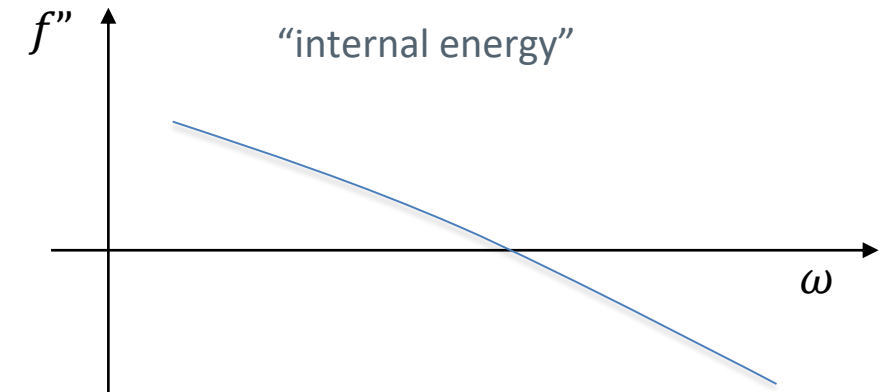
Legendre transforms

- Constrained electric displacement method (Stengel, Spaldin, Vanderbilt, Nat. Phys. 5, 304 (2009))

$$g(y) = \min_x \left[f(x) - yx \right] \quad y = \frac{df}{dx} \quad \longleftrightarrow \quad f(x) = \max_y \left[g(y) + yx \right] \quad x = -\frac{dg}{dy}$$



$$\frac{\partial^2 g}{\partial y^2} = -\left(\frac{\partial^2 f}{\partial x^2}\right)^{-1}$$



f and g contain the same information
 $f(\omega)$ more convenient for calculations

Thermodynamic Magnetic Functionals

Penalty-function constraints

Ma and Dudarev, Phys. Rev. B 91, 054420 (2015)

$$\tilde{U}(B_l) = E_{\text{KS}} + \frac{\alpha}{2} \sum_l [B_l - m_l]^2$$

α : Penalty amplitude

B_l : target magnetic moments

$$\tilde{F}(H_l) = E_{\text{KS}} - \sum_l H_l m_l - \sum_l \frac{H_l^2}{2\alpha}$$

H_l : local Zeeman fields on spheres (conjugate to B_l)

Holonomic constraints

Gonze et al., J. Chem. Theo. Comp. 18, 6099 (2022).

$$U(M_l) = E_{\text{KS}} + \sum_l H_l (M_l - m_l)$$

H_l : Lagrange multipliers

M_l : target magnetic moments

$$F(H_l) = E_{\text{KS}} - \sum_l H_l m_l$$

H_l : local Zeeman fields on spheres (conjugate to M_l)

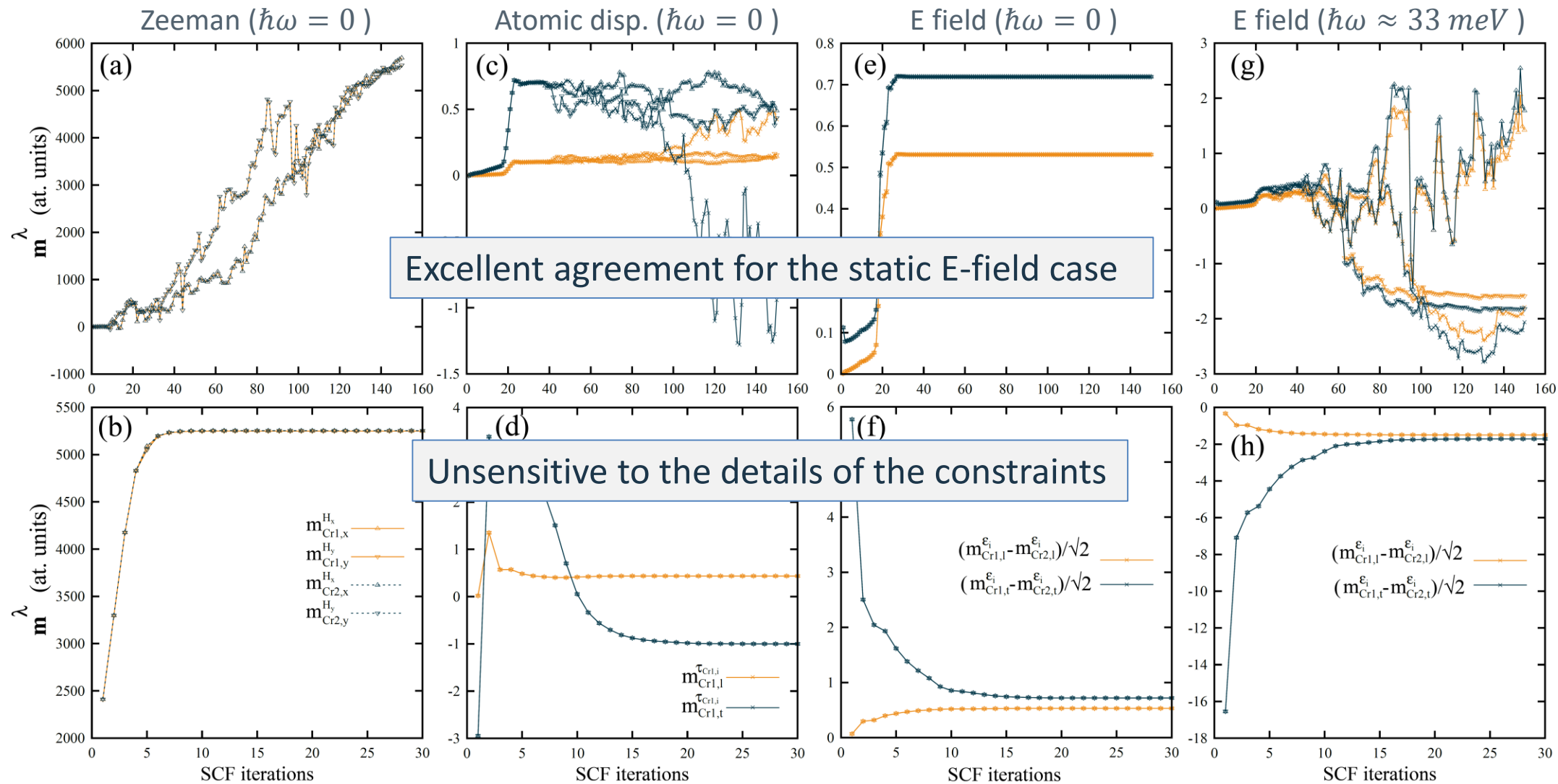
Can use second derivatives of $\tilde{U}$ or U to reconstruct the physical response F

$$\tilde{U}_{ab}(\omega) \rightarrow U_{ab}(\omega) \quad U_{ab}^{-1}(\omega) = -F_{ab}(\omega)$$

Does it work? Let's try again

DFPT

Constrained DFPT



Calculation of Spin Phonon Coupling Coefficients (Adiabatic Dynamics)

$$-\omega_n^2 M |q_n\rangle = -K |q_n\rangle - i\omega_n G |q_n\rangle$$

Ren, Bonini, Dreyer, Stengel,
Vanderbilt, PRX 2024

$$M = \underbrace{\begin{pmatrix} M^{(pp)} & 0 \\ 0 & 0 \end{pmatrix}}_{\text{Masses}} \quad K = \underbrace{\begin{pmatrix} K^{(pp)} & K^{(ps)} \\ K^{(sp)} & K^{(ss)} \end{pmatrix}}_{\text{Hessians}} \quad G = \underbrace{\begin{pmatrix} G^{(pp)} & G^{(ps)} \\ G^{(sp)} & G^{(ss)} \end{pmatrix}}_{\text{Berry Curvatures}}$$

- **Hessians:** 2nd-order derivatives at fixed spins (U_{ab})

$K^{(pp)}$: Interatomic force constants at fixed spins

$K^{(ss)}$: Local Zeeman fields induced by spin cantings at clamped ions

$K^{(sp)}$: Local Zeeman fields induced by ion displacements

$K^{(ps)}$: Forces induced by spin cantings

**DELIVERED WITH
UNPRECEDENTED
EFFICIENCY**

Calculation of Spin Phonon Coupling Coefficients (Adiabatic Dynamics)

- **Berry Curvatures:** Frequency derivative of U_{ab}

$$\mathbf{U}(\omega) = \mathbf{K} + i\omega\mathbf{G} - \omega^2\mathbf{M} + \dots$$

↖
↖
↖

Hessian Berry curvature Mass renormalization (electron inertia tensor)

$$\frac{dU_{ab}(\mathbf{q}, \omega)}{d\omega} = \frac{\partial U_{ab}(\mathbf{q}, \omega)}{\partial \omega} = \int [d^3k] \sum_m f_{n\mathbf{k}} (\langle u_{m\mathbf{k}, -\mathbf{q}}^b(-\omega) | u_{m\mathbf{k}, -\mathbf{q}}^a(-\omega) \rangle - \langle u_{m\mathbf{k}, \mathbf{q}}^a(\omega) | u_{m\mathbf{k}, \mathbf{q}}^b(\omega) \rangle)$$

- ✓ Calculation implemented in the longwave driver
- ✓ Stored as a 3rd-order total energy derivative in the DDB file
- ✓ Lineal interpolation in frequency of response functions (FOA) by ANADDB

Polynomial Frequency Interpolation of Response Functions

Local magnetic susceptibility (at $q=0$)

CONSTRAINED (resonance free)

$$\frac{\partial^2 U}{\partial M_j \partial M_l} = U_{jl} = \chi_{jl}^{-1}$$

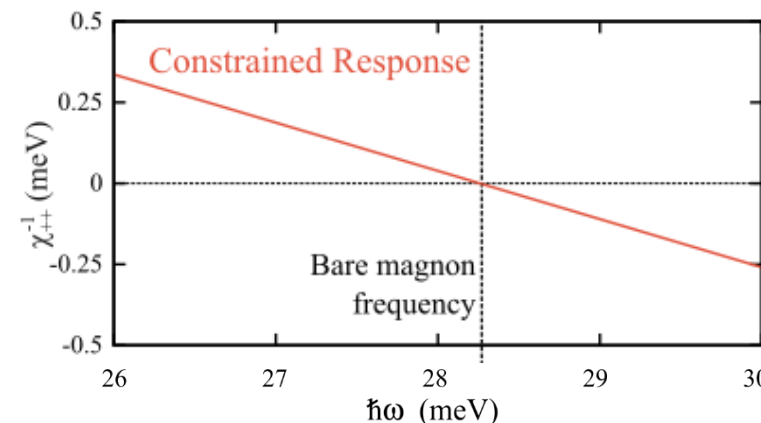
M_l : local magnetic moments

PHYSICAL (spin-relaxed)

$$\chi_{jl} = \left. \frac{\partial m_j}{\partial H_l} \right|_{F, \tilde{F}} = - \frac{\partial^2 F}{\partial H_j \partial H_l}$$

H_l : local Zeeman fields

First interpolate,
then relax



i) ABINIT:

- $\tilde{U}_{jl}(\omega)$ at N ω points
- N 2nd-order DDB files

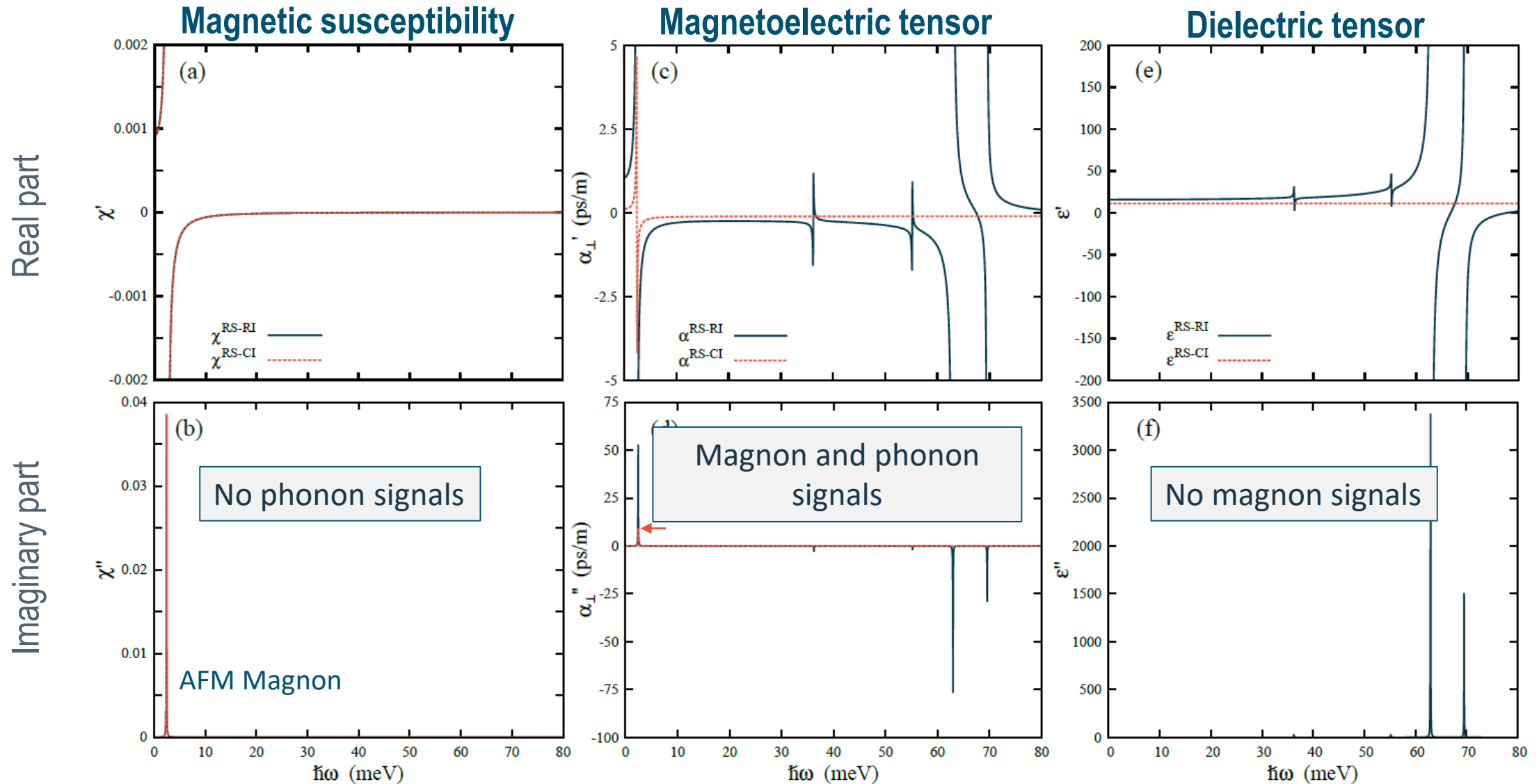
ii) MERGEDDB:

- Single DDB file with N blocks

iii) ANADDB:

- $\tilde{U}_{jl}(\omega_n) \rightarrow U_{jl}(\omega_n)$
- $N-1$ polynomial interpolation
- $U_{jl}(\omega) \rightarrow F_{jl}(\omega) \rightarrow \chi_{jl}(\omega)$

Frequency-Dependent Electromagnetic Linear Response of Cr_2O_3



Recap to This Point

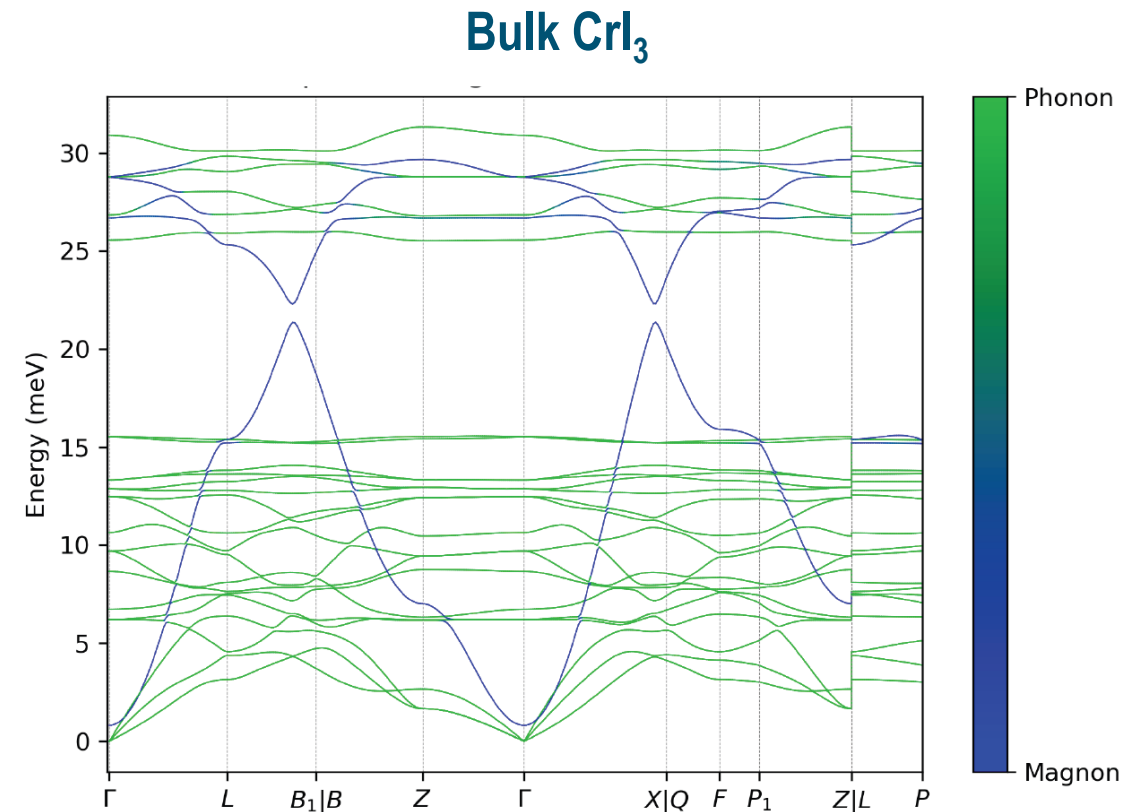
- Magnetic systems require to go beyond static Born-Oppenheimer
- Noncollinear TD-DFPT (and DFPT) challenging in magnets
- Constrained DFPT
 - Linear-response of magnets with comparable efficiency, accuracy and ease as in standard nonmagnetic insulators.
 - Nonadiabatic frequency-dependent electromagnetic response including phonons, magnons and hybrid modes.
 - Source of material coefficients (**K**, **G**, **M**) for models.
- Implemented and tested in ABINIT (merged in the trunk)

M. Royo and M. Stengel, PRX 16, 011049 (2026)



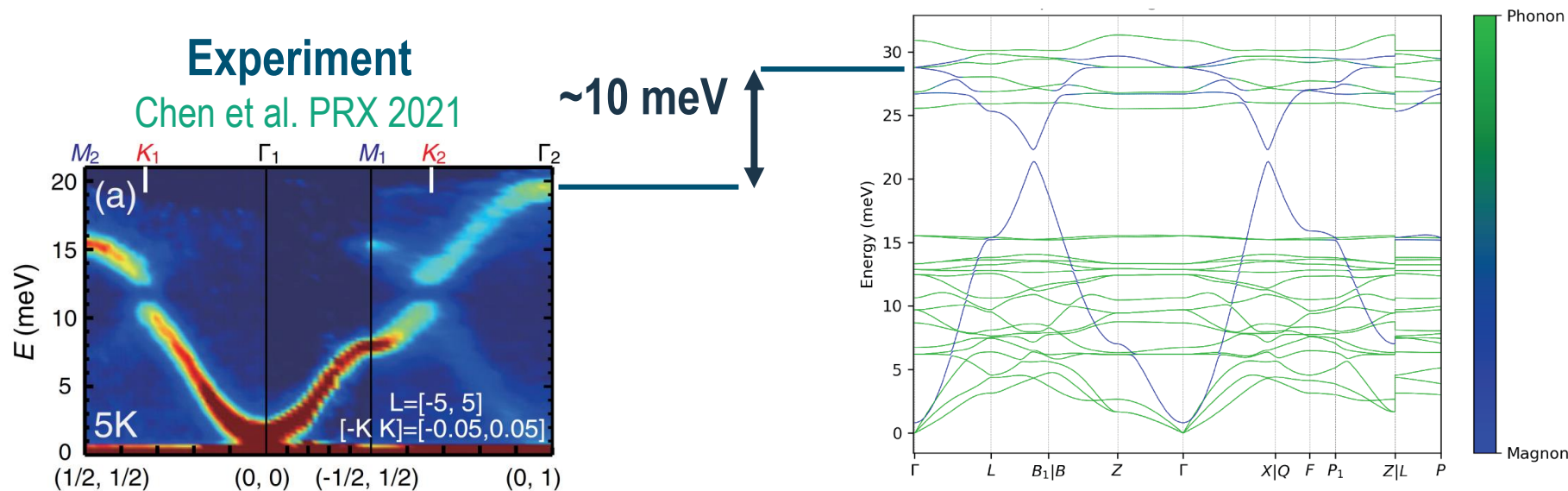
Brillouin-Zone Interpolation of Coupled Magnon-Phonon Bands

- ✓ Constrained DFPT at finite $\mathbf{q}$ and finite ω
- ✓ Berry Curvatures
- ✓ ANADDB post-processing
- ✗ $\mathbf{q}$ -interpolation of extended (spin-phonon) parameter space
- ✗ Treatment of nonanalytical contributions
- ✗ Generalized EOM solver



Collaboration with David Vanderbilt
(Rutgers) and Shang Ren (Berkeley)

Problem of Hard LDA Optical Magnons



- LDA and GGA yield too large spin stiffness ($K^{(ss)}$) coefficients (Gorni et al. PRB and PRM 2023)
- On-site Hubbard terms partially correct this (Ren et al. PRX 2024, Skovhus and Olsen PRB 2022)

Wish: LDA+U with NC PSPs in noncollinear DFT and DFPT drivers

Traveling Pseudopotentials Nonlocal Contributions with SOC

M. Stengel, M. Royo and E. Artacho, PRL 136, 196401 (2026)

“Velocity-dependent nonlocal operator → Restores the Galilean covariance and the identity between mechanical rototranslations and electromagnetic perturbations.”

✓ Implemented in the linear-response driver for nonmagnetic crystals (not yet in the trunk)

X Need to incorporate SOC

X Linear-in-frequency terms missing → Berry curvature breaks ASR

$$\hat{H}^{\tau_{\kappa\alpha}\tau_{\kappa'\beta}} = \delta_{\kappa\kappa'} \left(\frac{\partial^2 \hat{H}}{\partial R_{\kappa\alpha} \partial R_{\kappa\beta}} - i\omega \frac{\partial^2 \hat{V}_{\kappa}^{\text{nl}}}{\partial k_{\alpha} \partial R_{\kappa\beta}} + i\omega \frac{\partial^2 \hat{V}_{\kappa}^{\text{nl}}}{\partial R_{\kappa\alpha} \partial k_{\beta}} + \omega^2 \frac{\partial^2 \hat{V}_{\kappa}^{\text{nl}}}{\partial k_{\alpha} \partial k_{\beta}} \right)$$

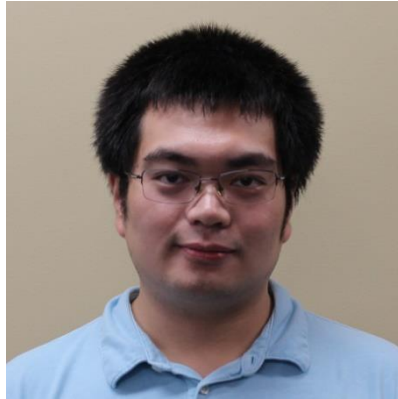
PROBLEM: Traveling PSP theory developed with Y_{lm} vs. DFPT with SOC implemented with P_l

Acknowledgments

Massimiliano Stengel



Shang Ren



David Vanderbilt



Samare Rostami



Xavier Gonze



Emilio Artacho



Thank you!



New Quantities in the DDB Files

Wave vector and frequency blocks

2nd derivatives (non-stat.) - # elements : 328

qpt 0.00000000E+00 0.00000000E+00 0.00000000E+00 1.0

frequency 6.00000000E-04

1 1 1 1 0.26233606350646D+02 0.000000000000000D+00

2 1 1 1 0.84376306520971D+01 -0.18946347061697D-09

...

3rd derivatives (long wave) - # elements : 7

qpt 5.00000000E-01 5.00000000E-01 0.00000000E+00 1.0

0.00000000E+00 0.00000000E+00 0.00000000E+00 1.0

0.00000000E+00 0.00000000E+00 0.00000000E+00 1.0

frequency 0.00000000E+00

0.00000000E+00

0.00000000E+00

1 1 1 1 1 15 -0.85265128291212D-13 0.000000000000000D+00

1 11 1 1 1 15 0.27455833366224D-07 0.15392282262158D-08

...

New ipert cases

Perturbation/Response

ipert= natom+5 → Macro. Zeeman field /magnetic moment

ipert= natom+6 → Scalar potential/charge

ipert= natom+11 : 2*natom+11 → Local Zeeman fields /magnetic moments

ipert= natom+9 → Frequency derivative